

Optogenetic Stimulation of Single Ganglion Cells in the Living Primate Fovea

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eLife Assessment

This paper shows that it is possible to optogenetically activate single retinal ganglion cells *in vivo* in monkeys. This is an **important** step towards towards causal tests of the role of specific ganglion cell types in visual perception. The paper presents **convincing** evidence for the promise of the approach but further work will be needed to full explore its limitations and specificity.

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Abstract

Though the responses of the rich variety of retinal ganglion cells (RGCs) reflect the totality of visual processing in the retina and provide the sole conduit for those processed responses to the brain, we have much to learn about how the brain uses these signals to guide behavior. An impediment to developing a comprehensive understanding of the role of retinal circuits in behavior is the paucity of causal studies in the intact primate visual system. Here we demonstrate the ability to optogenetically activate individual RGCs with flashes of light focused on single RGC somas *in vivo*, with no evidence for activation of neighboring cells. The ability to selectively activate specific cells is the first step toward causal experiments that directly link retinal circuits to visual experience and behavior.

Introduction

Though only a few classes of light-sensitive cells capture the retinal image in the primate eye, evolution has produced many more classes of retinal ganglion cell (RGC), on the order of 20, each morphologically and physiologically distinct and independently tiling the retina, to convey visual information to the brain. This dramatic expansion of cell classes in the retinal output combined with microelectrode recordings in excised tissue have made it increasingly clear that the diversity of RGCs reflects instead a surprising amount of complex early processing in the mammalian retina, processing that previously been had been thought to reside deeper in the brain (Gollisch and Meister, 2010 ). There is evidence, for example, for specialized circuits that convey

directional selectivity (DS) (Barlow et al., 1964 [↗](#)), segregate object from background motion (Lettvin et al., 1959 [↗](#)), and anticipate moving stimuli (Berry et al., 1999 [↗](#)). However, we remain uncertain about the direct functional role of these RGC classes because, while *ex vivo* recordings can characterize the specific stimulus requirements of RGC classes, they are often made in the excised retina that is detached from the downstream visual pathways they serve. Such studies preclude direct manipulations of RGC activity to observe the impact on visual experience or behavior in the awake primate. For example, we cannot say with certainty whether the DS cells recently observed in the non-human primate (NHP) are involved in eye movements, the perceptual experience of motion, or both. Indeed, there is even lingering controversy about the perceptual roles of the most common primate RGCs, the midget and parasol ganglion cells (Freedland and Rieke, 2022 [↗](#); Patterson et al., 2019 [↗](#)).

To address these controversies, one exciting possibility for the future is the development of studies in which the effect of manipulating RGC responses in the awake behaving animal can be measured with psychophysical or behavioral measures. There have been formidable challenges to deploying this causal paradigm to study retinal circuits because of their inaccessibility inside the moving eye of the awake behaving primate. Advances in high resolution retinal imaging with adaptive optics now make it possible to observe single RGCs in the living eye (Gray et al., 2008 [↗](#)). AO has also enabled very accurate eye tracking, making it possible to not only image but also stimulate single photoreceptors repeatedly with focused flashes of light (Schmidt et al., 2018a [↗](#),b; Tuten et al., 2017 [↗](#)) in the moving human eye. In addition, viral vectors now allow expression of calcium indicators such as GCaMP6s (Yin et al., 2013 [↗](#)a; Godat et al., 2022 [↗](#)), enabling recording from single RGCs *in vivo*. Finally, optogenetic activation of RGCs has been demonstrated in both macaque (McGregor et al., 2020 [↗](#)) and human (Sahel et al., 2021 [↗](#)) retina *in vivo*. These recent technical advances taken together now offer the possibility of causal experiments in which single RGCs are optogenetically stimulated in the living primate eye while the behavioral consequences are measured. Here, we demonstrate the first step toward establishing this capability: showing successful stimulation of single RGCs in the living primate eye.

Results and Discussion

RGC activity was modulated via the optogenetic ChrimsonR and monitored via fluorescence imaging of the calcium indicator GCaMP6s. Isolated foveal RGCs were chosen for individual stimulation, being targeted with short pulses of light intended to elicit activity in that cell only. The response of both the selected cell and all nearby cells were monitored to determine activation status. Chosen cells and their fluorescence traces can be seen in **Fig. 1** [↗](#). Targeted cells showed large responses to the optogenetic probe. The GCaMP responses rise to peak quickly, and then decay over several seconds. This time course is consistent with GCaMP responses to cone excitation of RGCs measured via *in vivo* AOSLO imaging (Yin et al., 2013b [↗](#)). **Fig. 2** [↗](#) shows all cells' $\Delta F/F$ values plotted against their distance from the targeted cell. Non-targeted cells showed no significant response above the background noise in contrast to the targeted cells, the responses of which were between 3 and 4 times the background noise floor. While we cannot exclude the possibility that surrounding cells responded to the flash below the noise threshold of our measurements, clearly these conditions strongly favor the excitation of the target cell over that of the neighbors we were able to record from. We cannot rule out the possibility that, due to the random nature of the vector expression process, some of these non-targeted cells were inactive because they failed to express ChrimsonR. The degree of GCaMP6s and ChrimsonR expression varies from one RGC to another, introducing the possibility that some cells may only have significant expression of one or neither agent. This means that some optogenetically active cells may have had no or weak GCaMP6s expression, diminishing our ability to observe their responses in our recordings. Despite this uncertainty, we can be reasonably sure that most cells are strongly co-expressing both agents. This is achieved via fluorescence imaging of the tdTomato marking ChrimsonR expression, which shows substantial overlap with the observed GCaMP6s fluorescence.

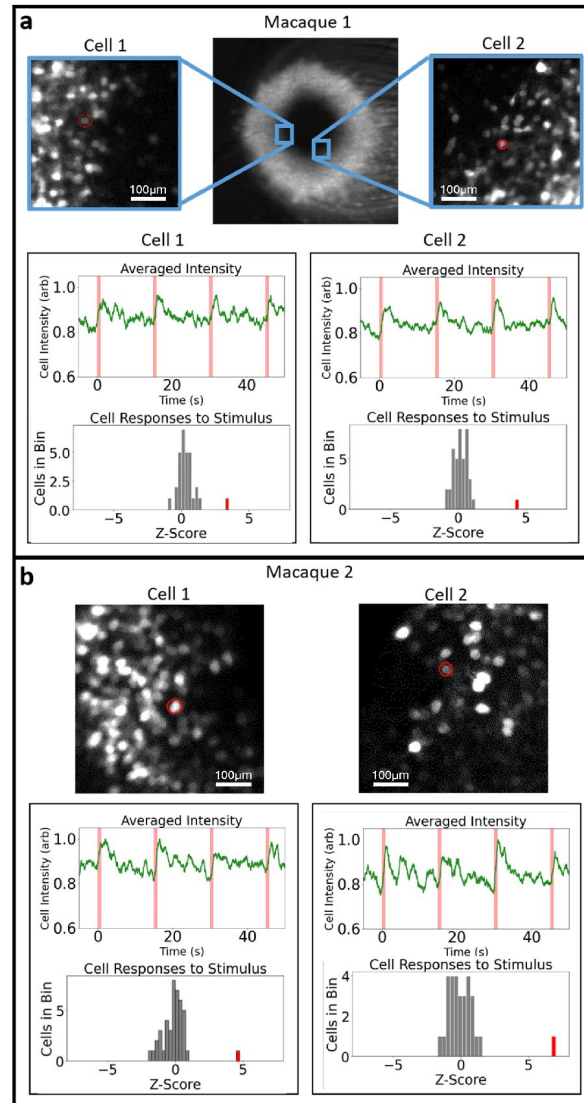


Figure 1.

Cells chosen for stimulation.

The cell circled in red marks the targeted cell. The averaged fluorescence traces for each cell show the integrated intensity over the cell's soma. The red marks on the time axis represent the onset of the 800 ms optogenetic stimulus. The histograms show each cell's response in terms of z score. (a) Results from male macaque. (b) Results from female macaque. All targeted cells are at similar eccentricities.

Figure 2.

$\Delta F/F$ for each cell soma plotted against that cell's distance from the targeted cell.

Note that even the nearby ($< 50 \mu\text{m}$) cell somas do not show a significantly elevated response ($p > 0.05$, unpaired t-test) than other cells at more distant locations. The leftmost point on each plot, colored in red, corresponds to the targeted cell itself. The traces for the four closest neighbors are shown below each scatter plot.

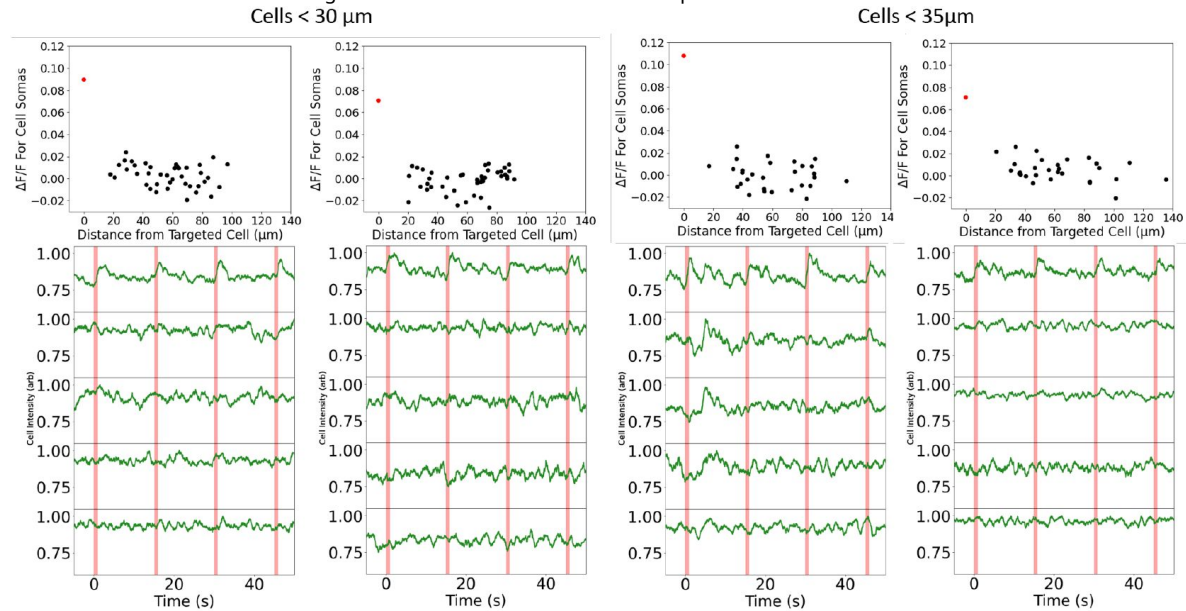
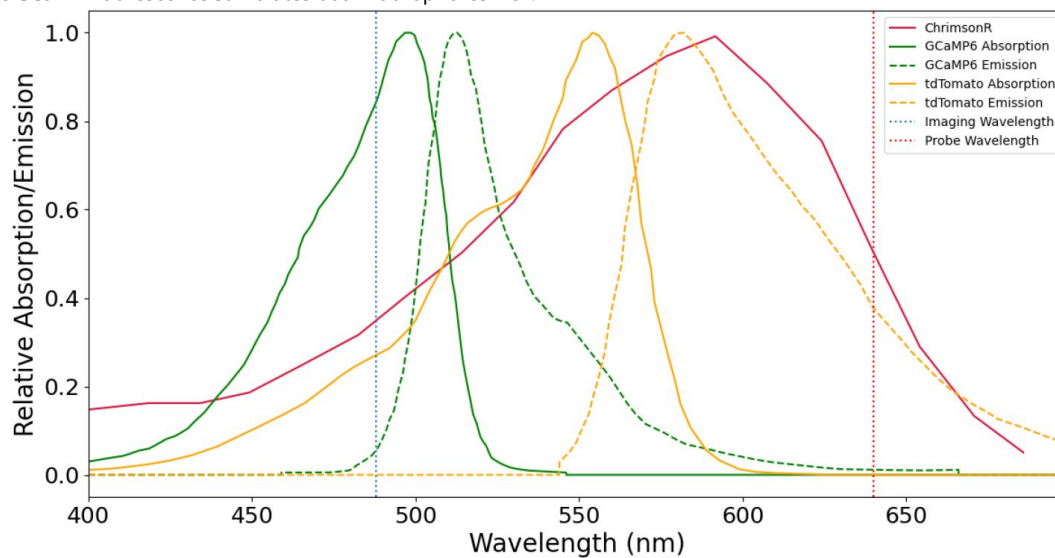


Figure 3.

Relevant absorption/emission spectra for our fluorophores and optogenetic actuator.

The emission wavelengths of tdTomato and GCaMP6 are well suited to activating ChrimsonR, and the wavelength used to image GCaMP fluorescence stimulates both fluorophores well.



Though this method has limitations, these may be overcome via further refinements of the technique. For this initial demonstration, we deliberately selected the most favorable conditions, especially by selecting cells to target that were spatially isolated from neighboring cells. Here, we show results for cells at the inner edge of the ring of RGCs serving the primate fovea. The ability to extrapolate the method to cells closely packed on all sides by neighboring cells, as occurs at slightly larger eccentricities, between 0.5 and 2.5 degrees, deeper in the foveal ring of labeled cells, has yet to be explored. In this region RGCs have little to no separation between cell somas. The transverse spatial resolution of an AOSLO is roughly 1.6 μm , which is more than adequate for targeting the much larger soma diameter. However, the axial resolution of an AOSLO is roughly 25 μm , roughly twice the soma diameter. This means that there would be some optogenetic excitation of the cells immediately above and below the cell of interest. But even in this less favorable case the balance of excitation could be shifted strongly to a subset of RGCs, even if perfect isolation were not possible. The ability to isolate single cells becomes far more favorable again in the periphery, where the RGC somas are in a monolayer. However, in this retinal region the RGC somas and their cone receptive fields lie on top of each other, precluding the use of the spatial offset between cone receptive fields and their somas to separate cone and RGC optogenetic excitation. The use of two photon excitation coupled with the use of a more sensitive optogenetic, such as ChRmine may mitigate this concern by increasing the optogenetic excitation while reducing the efficacy of light falling on photoreceptors (Marshall et al., 2019 [DOI](#)). Other approaches to increasing the selective activation of targeted cells may include cortically-targeted retrograde injections or class-specific viral vectors.

The demonstration of *in vivo* selective activation of individual RGCs lays the foundation for future causal studies of RGC function. The causal experimental approach has been successful in understanding the roles of neuronal classes in cortex of NHPs for many years, such as the work of Newsome and colleagues who showed that electrical activation of DS cells in MT modifies the animal's choices in motion discrimination tasks (Salzman et al., 1992 [DOI](#); Cicmil and Krug, 2015 [DOI](#); Parker and Newsome, 1998 [DOI](#)). More recently, this paradigm has incorporated optogenetic activation to investigate the function role of localized neural circuitry in cortex (Jazayeri et al., 2012 [DOI](#); Soma et al., 2019 [DOI](#)). Applied to RGCs, this approach would involve first, *in vivo* calcium imaging to classify each RGC, followed by psychophysical experiments combined with optogenetic stimulation designed to explore each cell's role in visual behavior. Such experiments would establish a direct link between activation of an individual RGC and visual behavior and could help determine whether a given cell class contributes to perceptual experience, subconscious visual behavior, or both. These experiments will face additional challenges such as deploying the technology demonstrated here in the awake behaving animal where eye motion will be larger than in the anesthetized animal. Nonetheless, the single cone tracking and stimulation experiments in the awake human inspire confidence that this will be possible (Schmidt et al., 2018a [DOI](#),b; Tuten et al., 2017 [DOI](#)). These experiments in which single cones are targeted with light have shown that visibility can be measured from photon absorption with flashes of light delivered through adaptive optics to single cones from each of the three foveal cone classes (Williams et al., 1981 [DOI](#); Schmidt et al., 2018 [DOI](#)b; Hofer et al., 2005 [DOI](#)), encouraging the possibility that optogenetic stimulation of a single RGC might be sufficient to elicit a sensation in an awake behaving monkey a hypothesis that future experiments could test. Monkeys are remarkably good at psychophysical experiments, with performance rivaling that of human observers (Matsuno and Fujita, 2009 [DOI](#)).

Interpretation of causal experiments through optogenetic stimulation of a single RGC is inevitably complicated by the fact that perceptual experience depends on cell populations and such experiments would necessarily produce unnatural population activity. How the brain would interpret such an unusual pattern of excitation is not entirely clear and it may be necessary to control the stimulation of multiple nearby RGCs to develop a clear understanding of their function. Additionally, should it prove possible to realize this paradigm, it could provide valuable data to guide optogenetics-based vision restoration therapies. This could eventually offer the opportunity

to simultaneously activate multiple classes of RGCs according to their individual response characteristics, leading to a perceptual experience that more closely resembles that generated by the natural environment in the sighted eye.

Methods

Animals and Animal Care

These experiments were performed in two anesthetized *macaca fascicularis*, one male (M1) and one female (M2). Macaques were housed in an AAALAC accredited facility and were provided with care by 4 full time veterinarians, 5 veterinary technicians, and an animal behaviorist from the Department of Comparative Medicine. The animal care staff monitored each animal for signs of dis-comfort at least twice daily. They were caged in pairs and had free access to lab chow and water. Their diet was additionally supplemented with daily treats including dried fruits, fresh fruits, vegetables, and nuts. All animals have daily access to enrichment items such as puzzle feeders and mirrors, and are provided with movies and/or music. More novel enrichment items such as treat filled bags, snow, or forage boxes were provided weekly. The macaques had access to a large play space with swings and perches on a rotating basis. This study was carried out in strict accordance with the Association for Research in Vision and Ophthalmology (ARVO) Statement for the Use of Animals and the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. One animal had two lesions of the central fovea, which had removed cone input to many of the labeled RGCs. The RGCs deafferented in this manner are easily distinguishable from those with normal input, and were avoided for study in this experiment.

Injections for Expression of Optogenetic and Calcium Indicator

Prior to injection, both animals received daily subcutaneous injections of cyclosporine to reduce the possibility of immune reaction to the injection. Blood trough levels were monitored to titrate the dose into the range of 150-200 ng/ml, and these levels were maintained throughout the testing period. The ocular surface was disinfected before injection with 50% diluted Betadine. Coexpression of both an optogenetic actuator (ChrimsonR) and a calcium indicator (GCaMP6s) enabled both stimulation of RGCs and recording of their responses. This was achieved with a single intravitreal injection containing a mixture of two adeno-associated virus (AAV2) based vectors with a ubiquitous CAG promoter, AAV2-CAG-tdTomato-ChrimsonR and AAV2-CAG-GCaMP6s. These vectors were synthesized by the University of Pennsylvania Vector core. The injection was delivered via a 30 gauge needle with a tuberculin syringe into the middle of the vitreous roughly 2 mm behind the limbus. The injection was made in the right eye of M1 and the left eye of M2, and had a total volume of 75 μ L. Post-injection, the eyes were monitored with conventional SLO imaging (Heidelberg Spectralis) for adverse events such as inflammation, which was not present in either animal throughout data collection. As previously demonstrated (Dalkara et al., 2013 [DOI](#)), expression was largely confined to an annulus spanning from 0.5° to 2°. The inner edge of this annular region corresponds to the most centrally located RGCs that receive input from the most central fovea, while the outer edge corresponds to the most eccentric location where we presume the inner limiting membrane remained thin enough for penetration of the viral vector. All RGCs that obtained expression are close enough to the foveal center to be displaced from their receptive fields, allowing for direct optogenetic activation of cell somas without stimulation of the cones that drive them.

AO Imaging

Anesthesia and Animal Preparation

Anesthesia and animal preparation were performed by licensed veterinary technicians. Macaques were fasted overnight prior to anesthesia in preparation for ventilation. The morning of an imaging session, macaques were moved to the laboratory and anesthetized with isoflurane. They were placed prone upon a stereotaxic cart, supported by cushions and covered with a warming blanket. The animals were paralyzed for the duration of imaging via an injection of vecuronium bromide. For the duration of paralysis, the animal's heart and respiratory rates, blood oxygenation levels, and electrocardiogram were monitored by a veterinary technician. The animal's pupils were dilated with drops of phenylephrine chloride (2.5%) and Tropicamide (1%), and the eye fitted with a rigid gas permeable contact lens to maintain corneal hydration. The animal's head was stabilized with ear bars and a chin rest, and the animal's whole body could be rotated via the stereotaxic cart to position the animal for imaging. The macaque was monitored for several hours by a veterinary technician after each session to ensure full and safe recovery from the anesthetic drugs. This procedure was repeated no more than once per week on an individual macaque.

Imaging Procedure

High resolution imaging was performed with an adaptive optics scanning light ophthalmoscope (AOSLO), a complete description of which can be found in (Godat et al., 2022 [DOI](#)). A Shack-Hartmann wavefront sensor in conjunction with an 847 nm laser diode source (QPhotonics) was used for wavefront sensing. The wavefront sensor data controlled a deformable mirror (ALPAO) in a closed-loop configuration operating at 10 Hz to provide correction of ocular aberrations. Reflectance and fluorescence signals were collected with photomultiplier tubes. The RGC layer was imaged using the fluorescent light emitted by the calcium indicator excited with light from a 488 nm laser (Qtop-tiq), and the optogenetic was excited using the 640 nm line of a multi-line laser (Toptica). The cone mosaic was simultaneously imaged with light from a 796 nm superluminescent diode (Superlum) to provide a higher SNR helpful for both real-time stimulus stabilization and post-recording image registration. Both uses are based upon cross-correlation of small strips of the image (Yang et al., 2014 [DOI](#)). For stimulus stabilization, only strips located near the stimulus' intended location were used in the cross correlation, allowing the stimulus to be repeatedly presented to the same retinal location. This stabilization is necessary because the heart beat and respiration of the animal cause lateral shifts in the retinal imaging throughout recording of approximately 50 μm . The image registration application is similar in execution and is applied in turn to all sections of the image to obtain a single stabilized video. The motion data from the strip registration of the cone reflectance video was used to stabilize the calcium recording. These methods allowed us to obtain cellular-scale recordings from the primate retina in vivo while correcting for the motion inherent to the living eye.

Cell selection

For this first study, we selected RGCs lying on the innermost edge of the annulus of expression often had large distances between their somas, making them easier to target with optogenetic stimulation without exciting neighboring cells. This crosstalk could potentially enter our measurements both when stimulating and recording. First, scatter from the 640 nm stimulus may illuminate nearby cell somas or underlying fibers of passage. Additionally, these nearby cell somas, their dendritic arbors, and fibers may contaminate the GCaMP recording if they overlap with the cell of interest's soma. We chose cells two cells in each of the two animals that were > 20 μm from their the nearest neighbor to mitigate the possibility of optical crosstalk. We have considered two possibilities by which light sources in the experiment other than the optogenetic probe might have interfered with the RGC response, but calculations confirm that neither would be expected to appreciably influence the probe response. First, the 488 nm imaging light is expected to produce an optogenetic activation that is only about 2% of that of the optogenetic probe. Moreover, it illuminates all cells equally throughout the experiment and would be expected

to produce a constant activation, unlike the probe which is confined in space and time. Second, the emission from GCaMP and tdTomato is at least 4 orders of magnitude lower than this 488 nm stimulus, making it unlikely that these fluorophores could have caused a measurable cell response.

Cells within a $1^\circ \times 1^\circ$ field of view were imaged at 488 nm (25 μ W) for 30 seconds to allow for adaptation to the maintained calcium imaging light. The RGCs showed some level of baseline activation in response to this. ChrimsonR was then activated by a circular stimulus (640 nm, 750 μ W) with a diameter of 12.5 μ m, approximately the size of a cell soma. This stimulus was formed by modulating the 640 nm laser source's output to only be on while scanning over the soma of interest at a rate of 25 Hz. The duration of excitation was 800 ms, throughout which the stimulus was stabilized on the cell's soma using the method describe in the imaging procedure section. This stimulus was repeated four times for each trial with a waiting period of 15 seconds between stimuli. We have seen no evidence that this modulated cell activity via the normal cone pathways.

Analysis

Analysis The GCaMP6s recordings were registered using strip registration and the frames averaged to obtain a high SNR image of the RGC mosaic. This registration is necessary because the heart beat and respiration of the animal cause lateral shifts in the retinal imaging throughout recording of approximately 50 μ m. The resulting RGC image was manually segmented to create a binary mask that can be applied to each frame of the video. The mask is used to extract the mean brightness of each cell soma for each frame. These brightness values are then used to calculate $\Delta F/F$ values for each cell soma in response to each stimulus. This calculation was performed for all cells that remain within the field of view for the entirety of the recording. Cells near the edge of the field of view often move in and out of visibility, obscuring any signals that may be present. The mean and standard deviation of $\Delta F/F$ values are calculated for periods during which no stimulus was present using data taken from all labeled RGCs in the subject's retina. These are used to transform the cell responses to stimuli into z-scores.

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Author Contributions

Peter J. Murphy, Conceptualization, Data curation, Formal analysis, Investigation, Software, Visualization, Methodology, Writing - original draft, Writing - reviewing and editing; Zhengyang Xu, Investigation, Methodology; Juliette E. McGregor, Investigation, Methodology, Writing - reviewing and editing; Qiang Yang, Software, Methodology; William Merigan, Conceptualization, Investigation, Methodology, Project administration, Supervision, Resources, Writing - reviewing and editing; David R. Williams, Conceptualization, Project administration, Supervision, Resources, Funding acquisition, Writing - original draft, Writing - reviewing and editing

Data Availability

All fluorescence recordings, along with example analysis scripts and required supporting files, are available at <https://osf.io/zgjsv/> 

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Reviewer #1 (Public review):

Summary

This manuscript reports preliminary evidence of successful optogenetic activation of single retinal ganglion cells (RGCs) through the eye of a living monkey using adaptive optics (AO).

Strengths

The eventual goals of this line of research have an enormous potential impact in that they will probe the perceptual impact of activating single RGCs. While I think more data should be included, the four examples shown look quite convincing.

Weaknesses

While this is undoubtedly a technical achievement and an important step along this group's stated goal to measure the perceptual consequences of single-RGC activations, the presentation lacks the rigor that I would expect from what is really a methods paper. In my view, it is perfectly reasonable to publish the details of a method before it has yielded any new biological insights, but in those publications, there is a higher burden to report the methodological details, full data sets, calibrations, and limitations of the method. There is considerable room for improvement in reporting those aspects. Specifically, more raw data should be shown for activations of neighboring RGCs to pinpoint the actual resolution of the technique, and more than two cells (one from each field of view) should be tested. Some information about the density of labeled RGCs in these animals would also be helpful to provide context for how many well-isolated target cells exist per animal.

<https://doi.org/10.7554/eLife.90050.2.sa3>

Reviewer #2 (Public review):

Murphy et al. expressed ChrimsonR and GCaMP6s in retinal ganglion cells of a living macaque. They recorded calcium responses and stimulated individual cells, optically. Neurons targeted for stimulation were activated strongly whereas neighboring neurons were not.

The ability to record from neuronal populations while simultaneously stimulating a subset in controlled way is a high priority for systems neuroscience, and this has been particularly challenging in primates. This study marks an important milestone in the journey towards this goal.

<https://doi.org/10.7554/eLife.90050.2.sa2>

Reviewer #3 (Public review):

This paper reports a considerable technical achievement: the optogenetic activation of single retinal ganglion cells in vivo in monkeys. As clearly specified in the paper, this is an important step towards causal tests of the role of specific ganglion cell types in visual perception. The paper is brief, and it will be important to follow this work with a more detailed methodological description to guide related work, to explore limitations, and to build confidence in the specificity of the approach.

<https://doi.org/10.7554/eLife.90050.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary

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Strengths

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We have expanded sections discussing both the methodology and limitations of this technique via a rewrite of the results and discussion section. The data used in the paper is available online via the link provided in the manuscript. We agree that a more detailed investigation of the strengths and limitations of the approach would have been a laudable goal. However, before returning to more detailed studies, we have shifted our effort to developing the monkey psychophysical performance we need to combine with the single cell stimulation approach described here. In addition, the optogenetic ChrimsonR used in this study is not the best choice for this experiment because of its poor sensitivity. We are currently exploring the use of ChRmine (as described in lines 93-97), which is roughly 2 orders of magnitude more sensitive. We have also been working on methods to improve probe stabilization to reduce tracking errors during eye movements. Once these improvements have been implemented, we will undertake the more detailed studies suggested here. Nonetheless, as a pragmatic matter, we submit that it is valuable to document proof-of-concept with this manuscript.

Some information about the density of labeled RGCs in these animals would also be helpful to provide context for how many well-isolated target cells exist per animal.

We agree. Getting reliable information about labeled cell density would be difficult without detailed histology of the retina, which we are reluctant to do because it would require sacrificing these precious and expensive monkeys from which we continue to get valuable information. We are actively exploring methods to reduce the cell density to make isolation easier including the use of the CAMKII promoter as well as the use of intracranial injections via AAV.retro that would allow calcium indicator expression in the peripheral retina where RGCs form a monolayer. It may be that the rarity of isolated RGCs will not be a fundamental limitation of the approach in the future.

Reviewer #2 (Public Review):

This proof-of-principle study lays important groundwork for future studies. Murphy et al. expressed ChrimsonR and GCaMP6s in retinal ganglion cells of a living macaque. They recorded calcium responses and stimulated individual cells, optically. Neurons targeted for stimulation were activated strongly whereas neighboring neurons were not.

The ability to record from neuronal populations while simultaneously stimulating a subset in a controlled way is a high priority for systems neuroscience, and this has been particularly challenging in primates. This study marks an important milestone in the journey towards this goal.

The ability to detect stimulation of single RGCs was presumably due to the smallness of the light spot and the sparsity of transduction. Can the authors comment on the importance of the latter factor for their results? Is it possible that the stimulation protocol activated neurons nearby the targeted neuron that did not express GCaMP? Is it possible that off-target neurons near the targeted neuron expressed GCaMP, and were activated, but too weakly to produce a detectable GCaMP signal? In general, simply knowing that off-target signals were undetectable is not enough; knowing something about the threshold for the detection of off-target signals under the conditions of this experiment is critical.

We agree with these points. We cannot rule out the possibility that some nearby cells were activated but we could not detect this because they did not express GCaMP. We also do not know whether cells responded but our recording methods were not sufficiently sensitive to detect them. A related limitation is that we do not know of course what the relationship is between the threshold for detection with calcium imaging and what the psychophysical detection threshold would have been in an awake behaving monkey. Nonetheless, the data show that we can produce a much larger response in the target cell than in nearby cells whose response we can measure, and we suggest that that is a valuable contribution even if we can't argue that the isolation is absolute. We've acknowledged these important limitations in the revised manuscript in lines 66-77.

Minor comments:

Did the lights used to stimulate and record from the retina excite RGCs via the normal lightsensing pathway? Were any such responses recorded? What was their magnitude?

The recording light does activate the normal light-sensing pathway to some extent, although it does not fall upon the RGC receptive fields directly. There was a 30 second adaptation period at the beginning of each trial to minimize the impact of this on the recording of optogenetically mediated responses, as described in lines 222-224. The optogenetic probe does not appear to significantly excite the cone pathway, and we do not see the expected off-target excitations that would result from this.

The data presented attest to a lack of crosstalk between targeted and neighboring cells. It is therefore surprising that lines 69-72 are dedicated to methods for "reducing the crosstalk problem". More information should be provided regarding the magnitude of this problem under the current protocol/instrumentation and the techniques that were used to circumvent it to obtain the data presented.

The "crosstalk problem" referred to in this quote refers to crosstalk caused by targeting cells at higher eccentricities that are more densely packed, which are not represented in the data. The data presented is limited to the more isolated central RGCs.

Optical crosstalk could be spatial or spectral. Laying out this distinction plainly could help the reader understand the issues quickly. The Methods indicate that cells were chosen on the basis that they were > 20 μm from their nearest (well-labeled) neighbor to mitigate optical crosstalk, but the following sentence is about spectral overlap.

We have added a clearer explanation of what precisely we mean by crosstalk in lines 213-221.

Figure 2 legend: "...even the nearby cell somas do not show significantly elevated response ($p > 0.05$, unpaired t-test) than other cells at more distant locations." This sentence does not indicate how some cells were classified as "nearby" whereas others were classified as being "at more distant locations". Perhaps a linear regression would be more appropriate than an unpaired t-test here.

The distinction here between "nearby" and "more distant" is 50 μm . We have clarified this in the figure caption. Performing a linear regression on cell response over distance shows a slight downward trend in two of the four cells shown here, but this trend does not reach the threshold of significance.

Line 56: "These recordings were... acquired earlier in the session where no stimulus was present." More information should be provided regarding the conditions under which this baseline was obtained. I assume that the ChrimsonR-activating light was off and the 488 nm GCaMP excitation light was on, but this was not stated explicitly. Were any other lights on (e.g. room lights or cone-imaging lights)? If there was no spatial component to the baseline measurement, "where" should be "when".

Your assumptions are correct. There was no spatial component to the baseline measurement, and these measurements are explained in more detail in lines 240-243.

Please add a scalebar to Figure 1a to facilitate comparison with Figure 2.

This has been done.

Lines 165-173: Was the 488 nm light static or 10 Hz-modulated? The text indicates that GCaMP was excited with a 488 nm light and data were acquired using a scanning light ophthalmoscope, but line 198 says that "the 488 nm imaging light provides a static stimulus".

The 488nm is effectively modulated at 25 Hz by the scanning action of the system. I believe the 10 Hz modulated you speak of is the closed-loop correction rate of the adaptive optics. The text has been updated in lines 217-219 to clarify this.

A potential application of this technology is for the study of visually guided behavior in awake macaques. This is an exciting prospect. With that in mind, a useful contribution of this report would be a frank discussion of the hurdles that remain for such application (in addition to eye movements, which are already discussed).

Lines 109-130 now offer an expanded discussion of this topic.

Reviewer #3 (Public Review):

This paper reports a considerable technical achievement: the optogenetic activation of single retinal ganglion cells in vivo in monkeys. As clearly specified in the paper, this is an important step towards causal tests of the role of specific ganglion cell types in visual

perception. Yet this methodological advance is not described currently in sufficient detail to replicate or evaluate. The paper could be improved substantially by including additional methodological details. Some specific suggestions follow.

The start of the results needs a paragraph or more to outline how you got to Figure 1. Figure 1 itself lacks scale bars, and it is unclear, for example, that the ganglion cells targeted are in the foveal slope.

The results have been rewritten with additional explanation of methodology and the location of the RGCs has been clarified.

The text mentions the potential difficulties targeting ganglion cells at larger eccentricities where the soma density increases. If this is something that you have tried it would be nice to include some of that data (whether or not selective activation was possible). Related to this point, it would be helpful to include a summary of the ganglion cell density in monkey retina.

This is not something we tried, as we knew that the axial resolution allowed by the monkey's eye would result in an axial PSF too large to only hit a single cell. The overall ganglion cell density is less relevant than the density of cells expressing ChrimsonR/GCaMP, which we only have limited info about without detailed histology.

Related to the point in the previous paragraph - do you have any experiments in which you systematically moved the stimulation spot away from the target ganglion cell to directly test the dependence of stimulation on distance? This would be a valuable addition to the paper.

We agree that this would have been a valuable addition to the paper, but we are reluctant to do them now. We are implementing an improved method to track the eye and a better optogenetic agent in an entirely new instrument, and we think that future experiments along these lines would be best done when those changes are completed.

The activity in Figure 1 recovers from activation very slowly - much more slowly than the light response of these cells, and much more slowly than the activity elicited in most optogenetic studies. Can you quantify this time course and comment on why it might be so slow?

We attribute the slow recovery to the calcium dynamics of the cell, and this slow recovery time is consistent with calcium responses seen in our lab elicited via the cone pathway. Similar time courses can be seen in Yin (2013) for RGCs excited via their cone inputs.

Traces from non-targeted cells should be shown in Figure 1 along with those of targeted cells.

We have added this as part of Figure 2.

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